ChakraA: how well the app hears a singing bowl

Run 2026-10-26_103325, chakra d7ddbb2, public variant

Summary

- **Strikes:** Flow shows the right chakra for 20 of 28 strikes (71 % [43, 81]), Strike for 21 of 28 (75 % [57, 81]); 2 recordings (YouTube only).
- **Latency** from the onset to the right colour, median: Flow 0.24 s [0.22, 0.25] (n = 8), Strike 0.11 s [0.05, 0.38] (n = 9).
- **The owner's own rounds** (seven bowls struck root to crown, 6 recordings, shown here as numbers only): Flow 38 of 42, Strike 40 of 42 strikes right.
- **AI gate** (mn05, 79 Commons clips): sounds that should colour keep 62 % [49, 74] without AI, 49 % [36, 62] pretrained, 47 % [34, 60] with the trained head; sounds that should not colour drop from 25 % [15, 36] to 6 % [2, 12] and 4 % [1, 9] (n = 30 and 49).
- **Light in real sessions** (10 YouTube sessions): Flow 7.5 [4.9, 8.2] colour changes a minute, Strike 15.0 [11.4, 19.1]; recordings with more than three changes in any second: Flow 0, Strike 2.

Every number carries its n. Intervals are 95 % percentile bootstrap intervals over recordings (10000 draws, seed 26102026): recordings, not strikes or windows, are the independent unit.

Data

Set	Recordings	Hours	What	Data list hash
Wikimedia Commons clips	79	0.82	free licences, 8 classes (fetch_clips.py)	`b89c100897ac`
YouTube videos	24	4.49	tune-ups, sessions on the body, concerts; not redistributed	`5d0a01aee0ea`
Own recordings	20	0.61	the owner's rounds and sessions; private, counted only	`957e45784bd5`

Left out as duplicates: CPIC2404IIs (the same audio as u1KjQTAHSIo under another channel, shifted by 1.7 s). Commons clips are single recordings of one instrument, not sessions; the YouTube and own sessions are the nearest thing to a room.

Data cleaning: 17 entries of the Commons source lists are recordings of someone saying the word ('shaker', 'gong', 'maraca': Lingua Libre and Wiktionary pronunciations), not the instrument; they are left out (fetch_clips.PRONUNCIATION). Up to 25 October 2026, 12 such files were counted among the benchmark's 91 clips (11 as shakers, 1 as a gong).

Method

- **Detector:** the app's own C code (`chakra_detect.c` through `tests/analyse_wav.c`, built into the run folder), notes at A = 440 Hz, sensitivity 8. **Flow** locks a tone after 384 ms and switches after 1500 ms; **Strike** after 128 and 256 ms. Both have the strike path (spectral flux) that lets a new bowl take over a louder ringing one.
- **Onsets** (reference, independent of the detector's decisions): peaks of the rectified log-magnitude spectral flux, 60 to 4000 Hz, 16 ms hops, corrected by 42 ms so a synthetic strike is found within 10 ms.
- **A strike is right** when its chakra is shown at some time from 0.1 s before the onset until the next strike or 3 s, whichever comes first. **First colour right**: the first colour that starts in that window is the right one. **Latency**: from the onset to the start of the right colour; a strike whose colour already showed (the same bowl again) counts for accuracy, not for latency.
- **Colour share:** the share of the sounding time (RMS above 3 % of the loudest 0.1 s) with any colour. The gate comparison uses gate_bench.py's own time base: the first 30 s of each clip.
- **Light:** colour changes a minute (a new chakra, or colour to dark), the share of colours that stand under 3 s, and the most changes in any 1 s window against the photosensitivity guidance of at most three flashes a second

(light_timeline.py).

- **AI gate:** EfficientAT mn05 on 2 s windows; pretrained = the best bowl-like AudioSet class beats the best other class; trained = a logistic head on the embedding, 5-fold by clip (a clip is never in its own training).

Ground truth and how far to trust it

Source	Chakra from	Strike time from	Reliability
Own rounds, hand-marked	a person marked each strike and its bowl (frames, or flux onsets checked against a contact sheet)	the flux onset found by the order alignment, when it lies within 1.6 s of the mark	chakra high; time high where the marks were checked (see validation)
Own rounds, derived	the seven bowls are struck root to crown (the owner's set, tuned C3 to B3)	onsets aligned to that order by dynamic programming: each bowl takes the onset where its fundamental gained most new energy	chakra high; time medium, validated on the round with checked marks
YouTube, chapters (-ar9vsmFhJU)	the player's chapter list, root to crown, 'each bowl is struck 3 times'	the 3 strongest onsets of each chapter, at least 4 s apart	chakra high; time medium
YouTube, announced (u1KjQTAHSlo)	automatic captions: the seller names each bowl's chakra and note	the strongest onset after the sentence and before the next one	chakra high; time medium
YouTube, crystal chapters (8Qhh2VF1Oz4)	chapter numbers read as chakra numbers, as the description says	none (bowls rung on): only the share of chapter time with the intended colour	medium

Validation of the derived strike times: the order alignment, run without the marks, against the hand marks of the two marked rounds.

Round	Strikes	Median offset	Largest	Within 0.1 s	Marks from
own round 1	7	1.12 s	1.53 s	0 of 7	marks from 1 fps frames
own round 2	7	0.03 s	0.04 s	7 of 7	checked marks

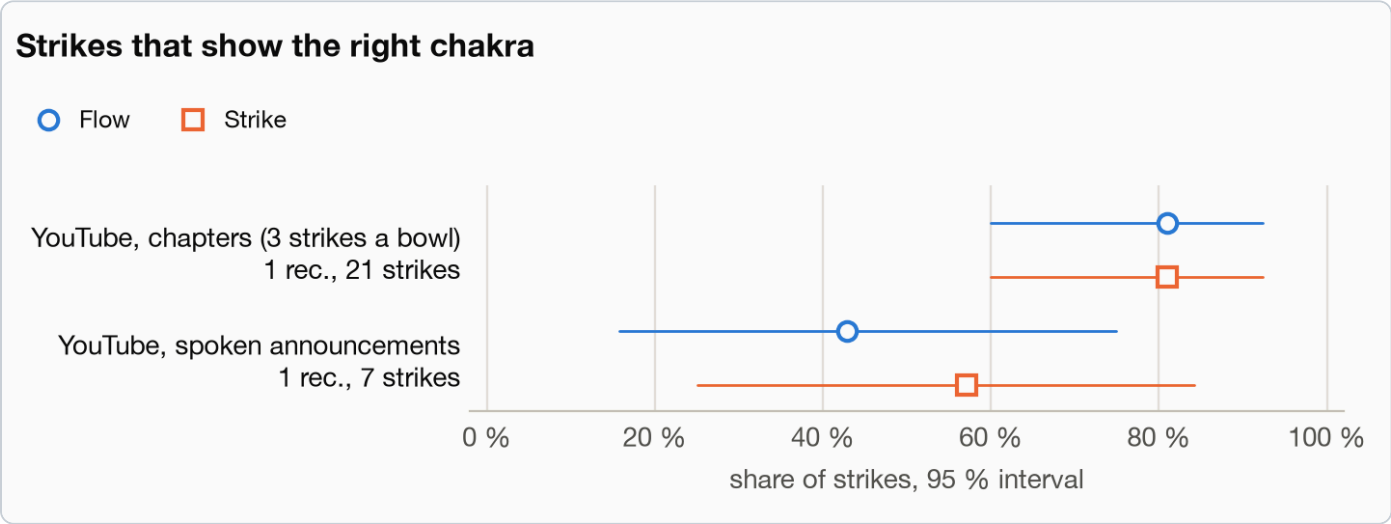
Where the marks were checked against the flux, the alignment finds the same strikes within a few hundredths of a second; where they came from video frames at one a second, the frames lag the sound by up to 1.5 s, so those marks give the bowl but not the time.

Results

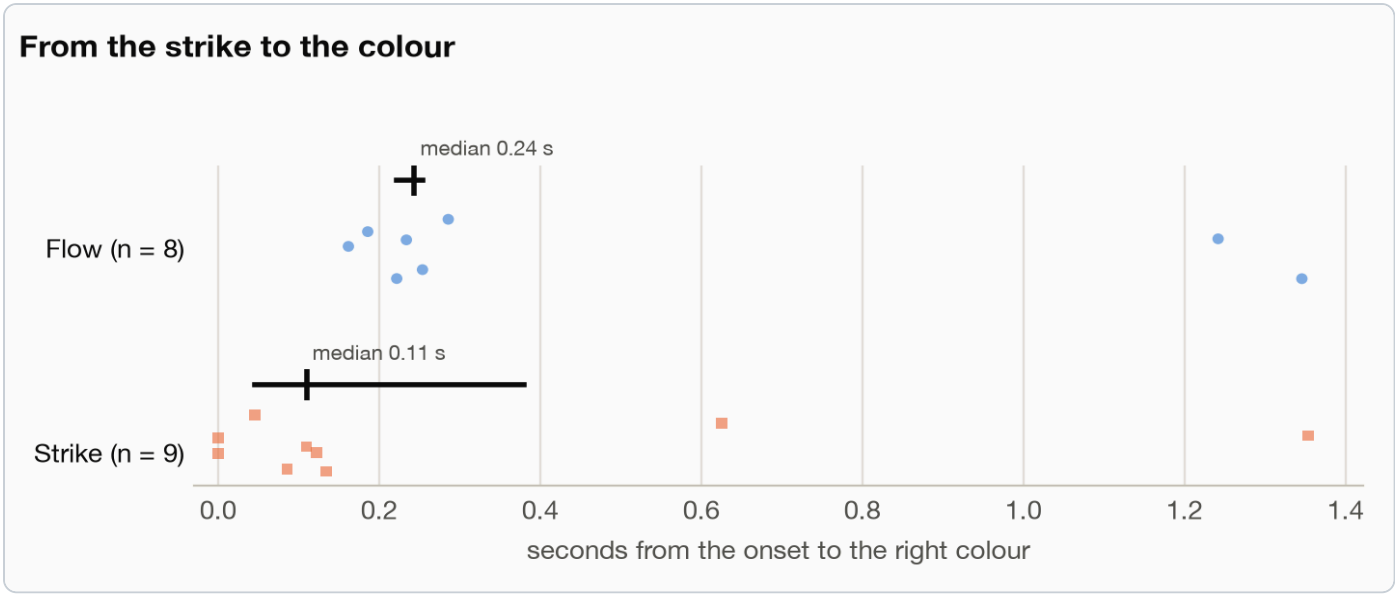
Strikes

Ground truth	Mode	Recordings	Strikes	Right	Accuracy [95 % CI]	First colour right	Latency, median [95 % CI]
YouTube, chapters (3 strikes a bowl)	Flow	1	21	17 of 21	81 %	81 %	0.22 s (n = 5)
YouTube, chapters (3 strikes a bowl)	Strike	1	21	17 of 21	81 %	76 %	0.05 s (n = 5)
YouTube, spoken announcements	Flow	1	7	3 of 7	43 %	43 %	0.25 s (n = 3)
YouTube, spoken announcements	Strike	1	7	4 of 7	57 %	43 %	0.38 s (n = 4)
Own rounds, hand-marked strikes	Flow	2	14	13 of 14	93 % [86, 100]	93 % [86, 100]	0.29 s [0.28, 0.33] (n = 13)
Own rounds, hand-marked strikes	Strike	2	14	13 of 14	93 % [86, 100]	93 % [86, 100]	0.01 s [0.01, 0.03] (n = 12)
Own rounds, derived strike times	Flow	4	28	25 of 28	89 % [86, 96]	89 % [86, 96]	0.30 s [0.29, 0.32] (n = 24)
Own rounds, derived strike times	Strike	4	28	27 of 28	96 % [89, 100]	93 % [86, 100]	0.02 s [0.01, 0.06] (n = 26)

Own rounds with derived times: 5 of 28 strikes are flagged uncertain (their bowl gained under 10 % of the new energy). Without them, Flow 91 % [83, 100], Strike 96 % [88, 100].



Share of strikes that show the right chakra, Flow and Strike. Filled marks and thick lines: 95 % bootstrap interval over recordings; hollow marks and thin lines: a group of one recording, Wilson interval over its strikes (YouTube; the own rounds are in the table).



Each point is one strike whose colour changed to the right chakra; the bar is the median with its 95 % bootstrap interval over recordings.

Time with the intended colour in the YouTube tune-ups, mean over their chapters or announced bowls:

Video	Truth	Segments	Flow	Strike
-ar9vsmFhJU	chapters	7	80 %	81 %
u1KjQTAHSlo	announced	7	54 %	51 %
8Qhh2VF1Oz4	chapters sustained	5	87 %	73 %

-ar9vsmFhJU's root bowl rings at G#3 (208 Hz): by note the app calls it Throat, so the root chapter scores near zero by design. The app's answer is to learn your bowls (Settings, mapping 'your bowls').

Colour by sound class

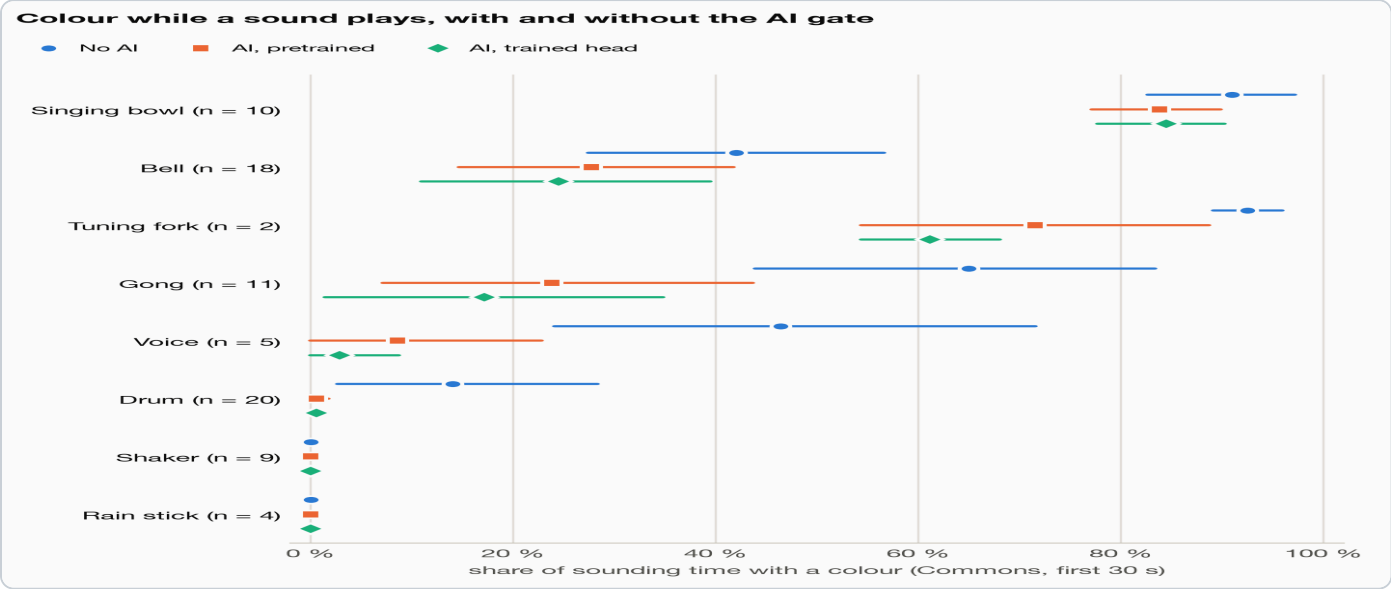
Sound	Recordings	Should colour	Flow, colour share	Strike, colour share
Singing bowl	10	yes	92 % [84, 97]	95 % [89, 99]
Bell	18	yes	43 % [30, 57]	85 % [77, 90]
Tuning fork	2	yes	91 % [87, 95]	95 % [93, 97]
Gong	11	no	56 % [35, 76]	93 % [88, 97]
Voice	5	no	46 % [24, 71]	81 % [68, 95]
Drum	20	no	15 % [3, 29]	46 % [28, 64]
Shaker	9	no	0 % [0, 0]	49 % [23, 73]
Rain stick	4	no	3 % [0, 8]	13 % [0, 38]

Mean share of the sounding time with any colour, whole recordings, 95 % bootstrap interval over recordings.

No AI, pretrained gate, trained head

Sound	Recordings	Should colour	No AI	AI, pretrained	AI, trained head
Singing bowl	10	yes	91 % [83, 97]	84 % [77, 90]	84 % [78, 90]
Bell	18	yes	42 % [27, 57]	28 % [15, 42]	24 % [11, 39]
Tuning fork	2	yes	93 % [89, 96]	72 % [54, 89]	61 % [54, 68]
Gong	11	no	65 % [44, 83]	24 % [7, 44]	17 % [1, 35]
Voice	5	no	46 % [24, 71]	9 % [0, 23]	3 % [0, 9]
Drum	20	no	14 % [3, 28]	1 % [0, 2]	1 % [0, 2]
Shaker	9	no	0 % [0, 0]	0 % [0, 0]	0 % [0, 0]
Rain stick	4	no	0 % [0, 0]	0 % [0, 0]	0 % [0, 0]

gate_bench.py, mn05_as@2s, 79 clips; the trained head gets 85 % of windows right on clips it never saw; 275 ms a window on the Mac's CPU.



Share of sounding time with a colour per class: no AI, the pretrained gate and the trained head.

The owner's recordings, segments a person checked (2 s labels), gate trained with them in the folds:

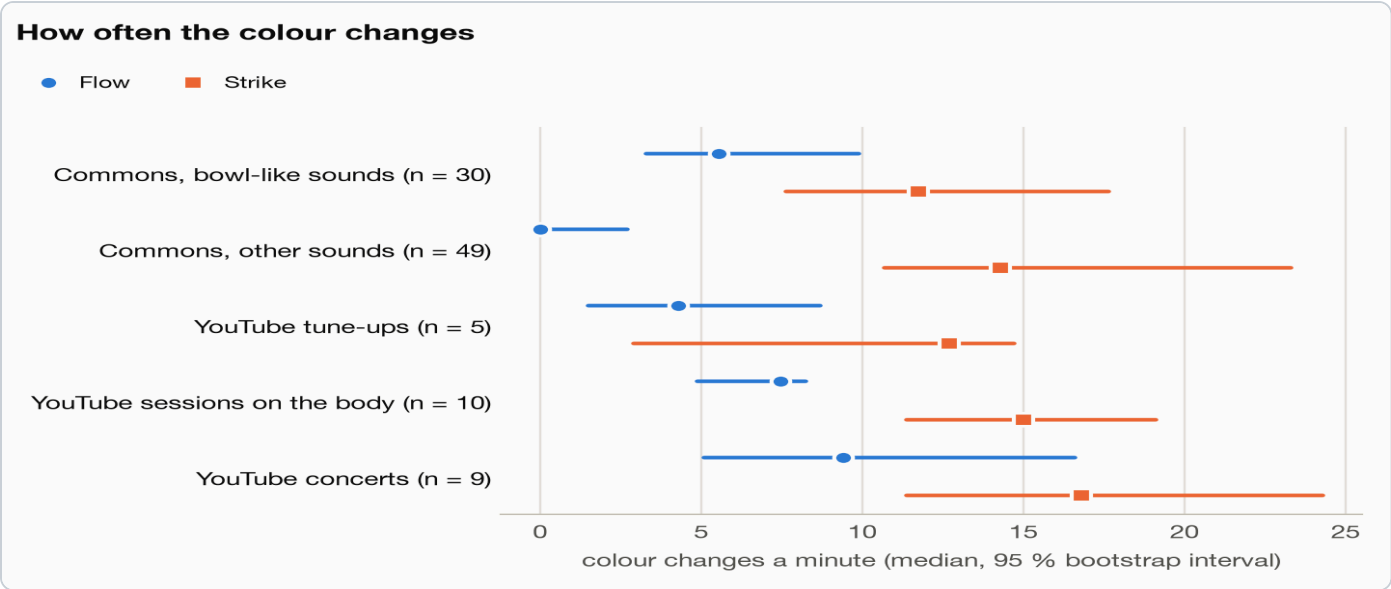
Segments	Recordings	No AI	AI, pretrained	AI, trained head
Own segments, bowl or bell (should colour)	10	69 % [53, 83]	41 % [23, 61]	63 % [47, 77]
Own segments, other sounds (should not)	3	10 % [0, 21]	0 % [0, 0]	7 % [0, 21]

Flow against Strike

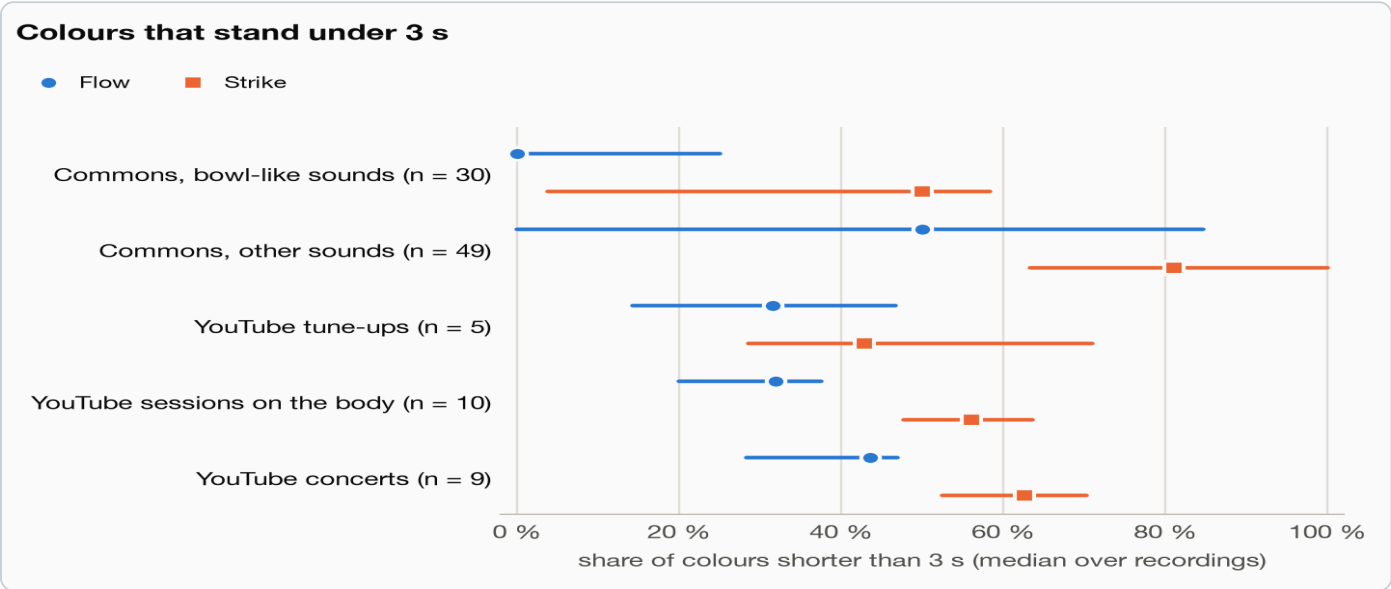
Measure	Flow	Strike
Strikes right (YouTube)	71 % [43, 81]	75 % [57, 81]
First colour right	71 % [43, 81]	68 % [43, 76]
Latency, median	0.24 s [0.22, 0.25]	0.11 s [0.05, 0.38]
Singing bowls, colour share	92 % [84, 97]	95 % [89, 99]
Gongs, colour share	56 % [35, 76]	93 % [88, 97]
Sessions, changes a minute	7.5 [4.9, 8.2]	15.0 [11.4, 19.1]
Concerts, changes a minute	9.4 [5.1, 16.6]	16.8 [11.4, 24.3]
Sessions, colours under 3 s	32 % [20, 38]	56 % [48, 64]

Light

Recordings	Mode	n	Colours	Changes a minute, median	Colours under 3 s, median	Over 3 changes in 1 s	Most changes in 1 s
Commons, bowl-like sounds	Flow	30	55	5.5 [3.3, 9.9]	0 % [0, 25]	0 of 30	2
Commons, bowl-like sounds	Strike	30	142	11.8 [7.7, 17.6]	50 % [4, 58]	0 of 30	3
Commons, other sounds	Flow	49	120	0.0 [0.0, 2.7]	50 % [0, 85]	0 of 49	2
Commons, other sounds	Strike	49	467	14.3 [10.7, 23.3]	81 % [63, 100]	1 of 49	4
YouTube tune-ups	Flow	5	104	4.3 [1.5, 8.7]	32 % [14, 47]	0 of 5	2
YouTube tune-ups	Strike	5	270	12.7 [2.9, 14.7]	43 % [29, 71]	0 of 5	3
YouTube sessions on the body	Flow	10	361	7.5 [4.9, 8.2]	32 % [20, 38]	0 of 10	3
YouTube sessions on the body	Strike	10	1479	15.0 [11.4, 19.1]	56 % [48, 64]	2 of 10	4
YouTube concerts	Flow	9	622	9.4 [5.1, 16.6]	44 % [28, 47]	0 of 9	3
YouTube concerts	Strike	9	1630	16.8 [11.4, 24.3]	63 % [52, 70]	1 of 9	4
Own rounds, derived strike times	Flow	6	47	8.9 [7.0, 14.5]	83 % [73, 94]	0 of 6	2
Own rounds, derived strike times	Strike	6	82	14.5 [11.8, 25.4]	80 % [65, 88]	0 of 6	3
Own sessions	Flow	14	114	8.9 [2.4, 12.3]	40 % [22, 65]	0 of 14	3
Own sessions	Strike	14	243	13.8 [6.8, 26.0]	71 % [60, 82]	0 of 14	3



Colour changes a minute per group of recordings, Flow and Strike: median and 95 % bootstrap interval over recordings.



Share of colours that stand under 3 s (median over recordings).

Limitations

- **Small n.** The strike results rest on 2 recordings; with so few, a bootstrap interval over recordings is coarse and can collapse to a point when every recording scores the same. Wilson intervals over strikes are in report.json; they assume independent strikes, which they are not.
- **Commons clips are not sessions.** They are short recordings of one instrument, often close up; a room with several bowls, voices and a gong at once is harder. The YouTube sessions and concerts have no ground truth and serve only the light measures.
- **Labels by one person.** The own segment labels, the strike marks and the check of the derived times were made by one person; nobody measured agreement between two.
- **Teacher-labelled data.** Unchecked labels of the own videos were proposed by an AudioSet model (EfficientAT dymn04); the gate's trained head uses only checked segments, but its pretrained teacher shaped what was checked. The YouTube chakra labels are what the players say their bowls are, not measured pitches.
- **One set of own bowls, one room.** All own rounds use the same seven bowls; another set, mallet or room may behave differently.
- **Latency resolution.** The detector prints its colour changes to 0.1 s, and the reference onset is the peak of the spectral flux, which for a soft felt mallet comes some tens of milliseconds after the attack begins. Latencies are good to about 0.1 s; a Strike latency near zero means 'within a hop or two of the flux peak', not 'before the sound'.

- **Not the board.** The detector runs on the Mac from audio files; the ESP32-S31's microphone, its gain and its timing are not measured here, and the lamps add their own delay.

Reproduce

- chakra (ChakraA) commit `d7ddbb2537bb` (Evaluation pipeline: stages from data to a scientific report, reproducible), 1 uncommitted paths at the start (untracked only)
- chakrav commit `78bfb56b88b0`
- stages continued later (--run latest --only report --force, --run latest --only report --force, --run latest --only report --force), last at chakra commit `9a35d3f92d27`
- config hash `c473ec5e2c85`; detector sources hash `4917f4824e88`; data hash `7b96ad157000`
- tools: Python 3.14.9, numpy 2.5.3, torch not in the pipeline interpreter (the gate stage runs in run.python), ffmpeg version 9.0.2 Copyright (c) 2000-2026 the FFmpeg developers
- command: `tools/bench/pipeline.py` (stages fetch, own, detect, strikes, gate, light, report); `--run latest --from report --force` redraws only the report.

Sources

YouTube (watched for evaluation only, nothing redistributed): -ar9vsmFhJU (templesounds); u1KjQTAHSlo (RANA SINGING BOWL CENTRE); ePKXtFLYnUs (templesounds); 8Qhh2VF1Oz4 (templesounds); _RijJiX5rng (templesounds); rVbdnUhjQhM; U4ximZISXWo; l1026mXyQol; PVNlmgsTbl0; HMClRBBC6iY; Tu3fBhpCusM; MDh00cYlmGk; WG9WGk55PG8; eXxitYpgNF0; lWPwzGgTmhc; BJKqpdvGiGc; vzMMYYl0K2k; VjJrvQCsc7o; tB1fvqGagFM; oh0FUveHOBI; chGb_m3wdZo; pqEG48dS6pw; 0sM7ggc3geU; 5EalbAG0lP8.

Wikimedia Commons: 79 clips under free licences, listed with title, page, licence and author in commons_sources.csv beside this report.

Per recording

Recording	Mode	Right	Strikes in order (+ right, x wrong)
-ar9vsmFhJU	Flow	17 of 21	xRoot xRoot xRoot +Sacr +Sacr +Sacr xSola +Sola +Sola +Hear +Hear +Hear +Thro +Thro +Thro +Thir +Thir +Thir +Crow +Crow +Crow
-ar9vsmFhJU	Strike	17 of 21	xRoot xRoot xRoot +Sacr +Sacr +Sacr xSola +Sola +Sola +Hear +Hear +Hear +Thro +Thro +Thro +Thir +Thir +Thir +Crow +Crow +Crow
u1KjQTAHSlo	Flow	3 of 7	+Crow +Thir xThro xHear xSola +Sacr xRoot
u1KjQTAHSlo	Strike	4 of 7	+Crow +Thir xThro +Hear +Sola xSacr xRoot